

综合练习

人员

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上周作业检查

上上周作业链接: <https://vjudge.net/contest/805933>

开始时间 : 2026-04-19 08:30 CST

☆ 2026-0419 六队上课 (综合练习)

结束时间 : 2026-11-13 16:30 CST

已开始 : 16:05:47:27

进行中

还剩余 : 192:02:12:32

概览 题目 提交状态 排名 (16:05:47:27) 讨论

设置 克隆 更新 删除

Rank	Team	Score	Penalty	A 7 / 9	B 7 / 7	C 7 / 19	D 7 / 19	E 3 / 3
1	☆ B_neutrino (Vm1m3)	5	55259	7:11:42:26 (-1)	7:10:43:10	7:12:29:14 (-3)	7:06:10:02 (-2)	8:13:54:29
2	☆ junyan007008	5	74984	7:01:01:00	13:01:26:00	11:13:33:00 (-1)	7:00:39:00	13:08:45:00
3	☆ lyc12345bc (刘奕辰)	5	88354	7:05:54:50 (-1)	15:13:44:00	7:06:52:00 (-4)	15:13:46:00 (-1)	15:14:18:00
4	☆ yycj	4	33882	1:39:37	9:07:22:58	7:01:44:49 (-1)	7:00:55:13 (-2)	
5	☆ r111	4	48791	8:13:13:00	9:13:43:00	8:12:01:00	7:06:14:00	
6	☆ ljjinshu	4	78526	15:13:18:21	15:22:14:00	15:22:22:00 (-2)	7:01:12:00 (-3)	
7	☆ wcz123bc	4	87432	15:03:29:00	15:04:37:00	15:04:18:00 (-1)	15:03:48:00 (-2)	
8	☆ 7654Fox (隋天乙)	0	0				(-2) +1	

上周作业链接: <https://vjudge.net/contest/807399>

本周作业

<https://vjudge.net/contest/810246> (课上讲了上周的 A ~ E 题, 课后作业是本周的 A ~ E 题)

课堂表现

今天的题目除了 A 题都相对难写一些, 需要同学们课后好好补一补。

课堂内容

CF1485D Multiples and Power Differences

用 720720 和 $720720+a[i][j]^4$ 来交替构造这个二维数组即可

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 500 + 5;
```

```

int w[maxn][maxn];

int main()
{
    int n, m; cin >> n >> m;
    for (int i = 1; i <= n; ++i) {
        for (int j = 1; j <= m; ++j) cin >> w[i][j];
    }

    for (int i = 1; i <= n; ++i) {
        for (int j = 1; j <= m; ++j) {
            if ((i+j)&1) cout << 720720 << " ";
            else cout << 720720 - w[i][j]*w[i][j]*w[i][j]*w[i][j] << " ";
        }
        cout << endl;
    }
    return 0;
}

```

CF1473D Program

输入 [l,r] 后, 可以查询 [1,l-1] 和 [r+1,n] 的最大值、最小值, 然后处理

```

#include <bits/stdc++.h>

using namespace std;

const int maxn = 2e5 + 5;
char s[maxn];
int _log2[maxn], p[maxn], f1[maxn][20], f2[maxn][20];
int n, m;

int get_max(int l, int r) {
    int len = r - l + 1;
    int k = _log2[len];
    return max(f1[l][k], f1[r-(1<<k)+1][k]);
}

int get_min(int l, int r) {
    int len = r - l + 1;
    int k = _log2[len];
    return min(f2[l][k], f2[r-(1<<k)+1][k]);
}

int calc(int l, int r) {
    int up = 0, down = 0;
    if (l-1 >= 1) up = max(up, get_max(1, l-1)), down = min(down, get_min(1, l-1));

    if (r+1 <= n) {
        up = max(up, get_max(r+1, n) - p[r] + p[l-1]);
        down = min(down, get_min(r+1, n) - p[r] + p[l-1]);
    }
}

```

```

    }
    return up - down + 1;
}

void solve() {
    cin >> n >> m;
    cin >> (s+1);
    for (int i = 1; i <= n; ++i) {
        if (s[i] == '+') p[i] = p[i-1] + 1;
        else p[i] = p[i-1] - 1;
        f1[i][0] = f2[i][0] = p[i];
    }

    for (int k = 1; k < 20; ++k) {
        for (int i = 1; i+(1<<k)-1 <= n; ++i) {
            f1[i][k] = max(f1[i][k-1], f1[i+(1<<(k-1))][k-1]);
            f2[i][k] = min(f2[i][k-1], f2[i+(1<<(k-1))][k-1]);
        }
    }

    while (m -- ) {
        int l, r; cin >> l >> r;
        // cout << "----- ";
        cout << calc(l, r) << endl;
    }
}

int main()
{
    for (int i = 0; (1<<i)<maxn; ++i) _log2[1<<i] = i;
    for (int i = 1; i < maxn; ++i) {
        if (_log2[i]) continue;
        _log2[i] = _log2[i-1];
    }

    int T; cin >> T;
    while (T -- ) solve();
    return 0;
}

```

CF981D Bookshelves

从二进制高位 枚举到 二进制低位, 然后再用 dp 判断能否把这一位变成 1 即可

```

#include <bits/stdc++.h>

using namespace std;

typedef long long LL;
const int maxn = 50 + 5;
LL w[maxn], p[maxn];

```

```

bool f[maxn][maxn];
int n, m;

LL get_sum(int l, int r) { return p[r] - p[l-1]; }

bool check(LL x) {
    memset(f, false, sizeof(f));
    f[0][0] = true;
    for (int i = 1; i <= n; ++i) {
        for (int j = 1; j <= m; ++j) {
            for (int k = 0; k <= i-1; ++k) {
                if (f[k][j-1] && (get_sum(k+1,i)&x)==x) f[i][j] = true;
            }
        }
    }
    return f[n][m];
}

int main()
{
    cin >> n >> m;
    for (int i = 1; i <= n; ++i) cin >> w[i], p[i] = p[i-1] + w[i];

    LL res = 0;
    for (int i = 60; i >= 0; --i) {
        if (check(res+(1LL<<i))) res += (1LL<<i);
    }
    cout << res << endl;
    return 0;
}

```

CF366C Dima and Salad

要求 $a/b == k$, 则可以转化为 $a - k*b == 0$, 然后做 01 背包找最大值即可

```

#include <bits/stdc++.h>

using namespace std;

const int N = 10000 + 5;
int a[N], b[N], p[2*N], f[2*N];

int main()
{
    int n, k; cin >> n >> k;
    for (int i = 1; i <= n; ++i) cin >> a[i];
    for (int i = 1; i <= n; ++i) cin >> b[i];

    memset(p, -0x3f, sizeof(p)); p[N] = 0;
    memset(f, -0x3f, sizeof(f)); f[N] = 0;
    for (int i = 1; i <= n; ++i) {

```

```

    int x = a[i] - k*b[i];
    for (int j = 0; j < 2*N; j++) {
        if (j-x>=0 && j-x<2*N) f[j] = max(f[j], p[j-x] + a[i]);
    }
    memcpy(p, f, sizeof(p));
}

cout << (f[N]>0 ? f[N] : -1) << endl;
return 0;
}

```

CF417D Cunning Gena

先按照 k 从小到大排序, 考虑以每个 k 进行结尾时, 做 状压dp 处理即可

```

#include <bits/stdc++.h>

using namespace std;

typedef long long LL;
const int N = 100 + 5, M = 21;
const LL inf = 0x3f3f3f3f3f3f3f3f;
struct node {
    int x, k, value;
    bool operator < (const node& p) const { return k < p.k; }
} w[N];
LL f[1<<M];

int main()
{
    int n, m, b; cin >> n >> m >> b;
    for (int i = 1; i <= n; ++i) {
        int x, k, m, value = 0; cin >> x >> k >> m;
        while (m -- ) { int c; cin >> c; --c; value |= (1<<c); }
        w[i] = {x, k, value};
    }
    sort(w+1, w+n+1);

    LL res = inf;
    memset(f, 0x3f, sizeof(f)), f[0] = 0;
    for (int i = 1; i <= n; ++i) {
        int x = w[i].x, k = w[i].k, value = w[i].value;
        for (int j = (1<<m)-1; j >= 0; --j) f[j|value] = min(f[j|value], f[j]+x);
        res = min(res, f[(1<<m)-1]+(LL)b*k);
    }

    cout << (res==inf ? -1 : res) << endl;
    return 0;
}

```